

ECSP3000 Project Brief

*Designing and Implementing an Automated & Reproducible
Testbed for Linux Rootkit Detectors, Aiming to Identify and
Implement a Novel Rootkit Technique*

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September 28, 2026

1. Project Aims

Current research into the evaluation of rootkits and rootkit detectors rely on bespoke evaluation methodologies which cannot be reliably or easily reproduced by independent parties.

Furthermore, as the Linux kernel gets updated with new features & security measures, it changes the landscape for these rootkits & rootkit detectors, creating a need for a consistent way to evaluate them continuously.

This tool aims to provide developers, researchers and system administrators with an automated and reproducible testbed for rootkits & rootkit detectors on the modern Linux kernel to aid future development and testing.

As an extension, I will use this tool to inform design of a novel proof-of-concept rootkit implementation for the modern Linux kernel, completing an evaluation of the PoC using the testbed.

2. Project Objectives

This project will have the following aims which will be used to gauge success:

1. To have a testing framework which, upon specifying a set of rootkits & rootkit detectors, provides timely & accurate results which can then be reproduced exactly upon rerun.
2. The system will aim to treat rootkits and rootkit detectors as pluggable modules, allowing developers to easily test new rootkit detectors or test against new rootkits.
3. If found to be feasible, to research and develop a proof-of-concept rootkit for the modern Linux kernel, following with an evaluation using this testing framework.
4. Using a dataset of results produced by the testbed or, if successful, the aforementioned PoC, evaluate the current state of rootkit detection technologies on the modern Linux Kernel.

3. Limitations & Scope

In order to scope my project appropriately, here are a list of things that I currently plan on **not** doing.

1. My testbed and therefore my rootkit research will only be concerned with kernel-space & user-space rootkits. This excludes boot-level, hypervisor-based or firmware- and hardware-level rootkits from the testbed's scope.
2. Testing rootkits that implement anti-sandbox measures fall out of scope for this project due to the virtualised environment employed by the testing framework.
3. "Modern Linux kernel" refers to the 7.x and 6.x series where the exact version(s) is to be determined conditional on software compatibility and time constraints.
4. The rootkit detectors that will be researched in this project will be on-host only, excluding memory forensics, hypervisor-based techniques or any hardware based detection methods from being in scope for my testbed.